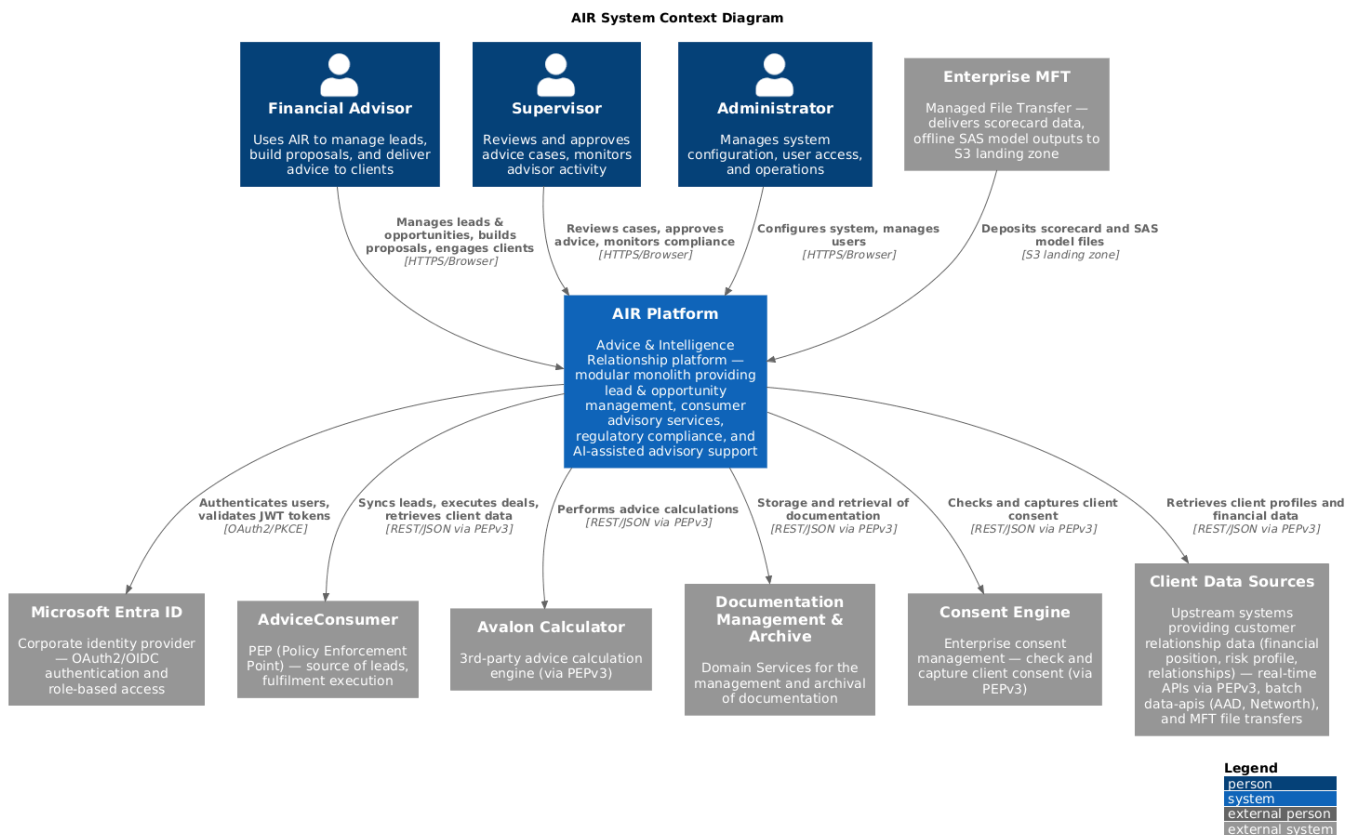


AIR (Advice & Intelligence Relationship) System Architecture Design – V2.3

Version: 2.3
Based on: System Architecture Design V2.2 + ADR-036 (Audit Trail) + ADR-037 (UI Theming)
Date: June 2026

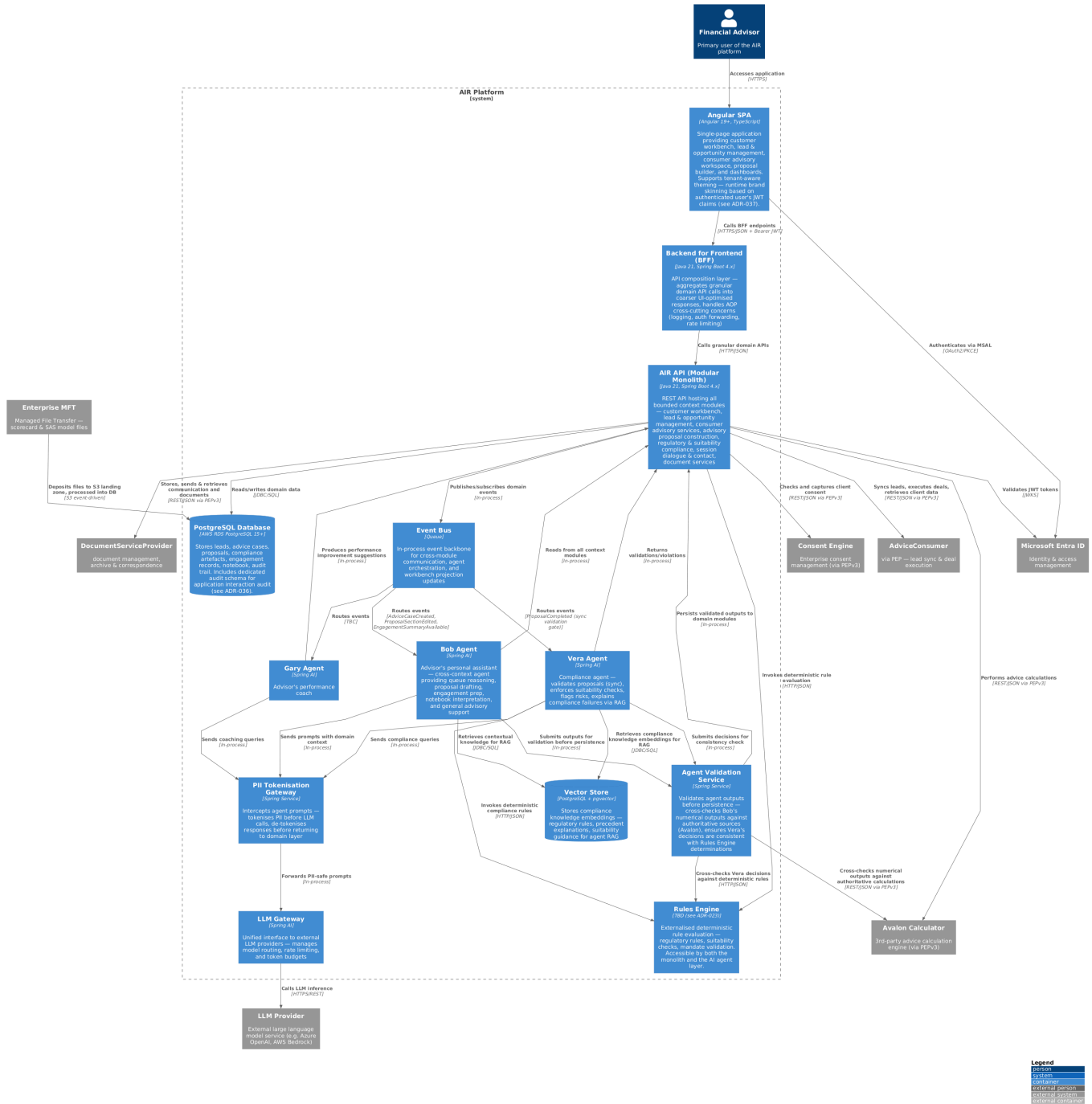
Level 1: System Context Diagram

Shows the AIR system in its environment, with external actors and systems.



Level 2: Container Diagram

Shows the major deployable units and their interactions.



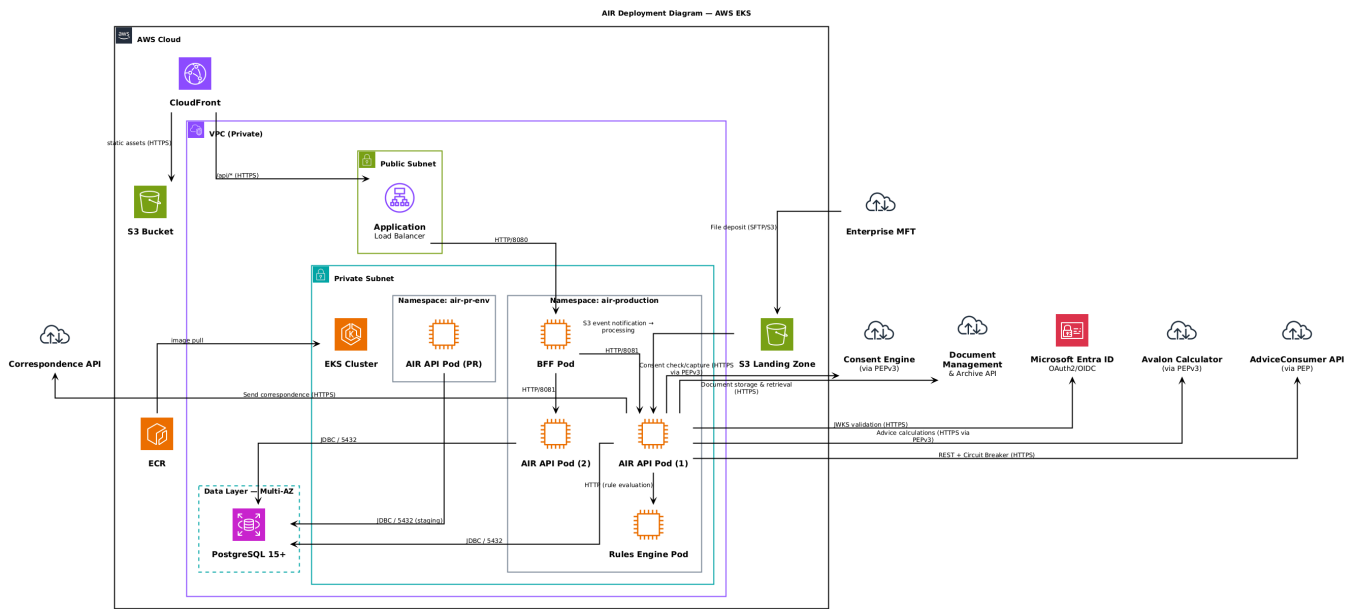
Level 3: Component Diagram – AIR API (Modular Monolith)

Shows the internal module structure of the monolith and its bounded contexts.



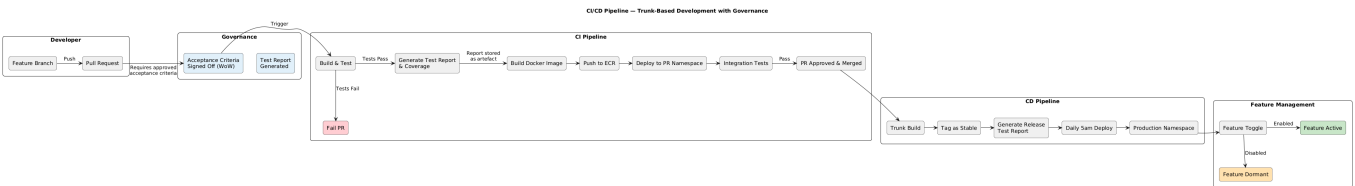
Level 4: Deployment Diagram

Shows the physical deployment topology on AWS. Deployment is infrastructure-level and independent of domain naming.



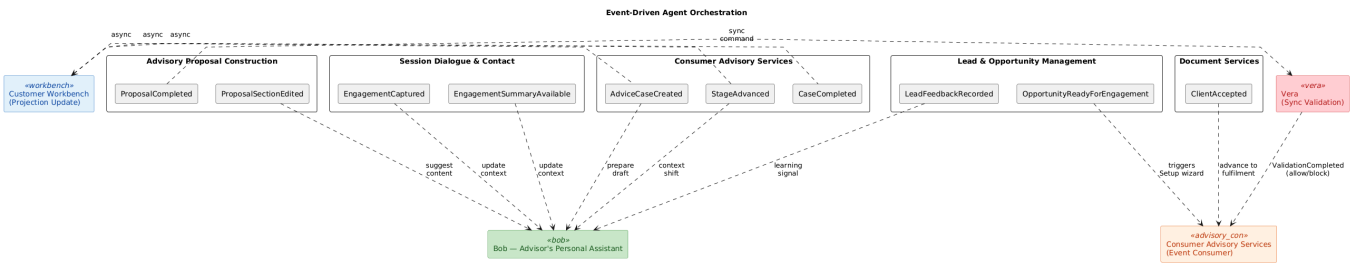
CI/CD Pipeline Flow

Illustrates the trunk-based development and deployment pipeline with governance gates.



Event Flow Diagram

Shows the event-driven orchestration pattern across bounded contexts.



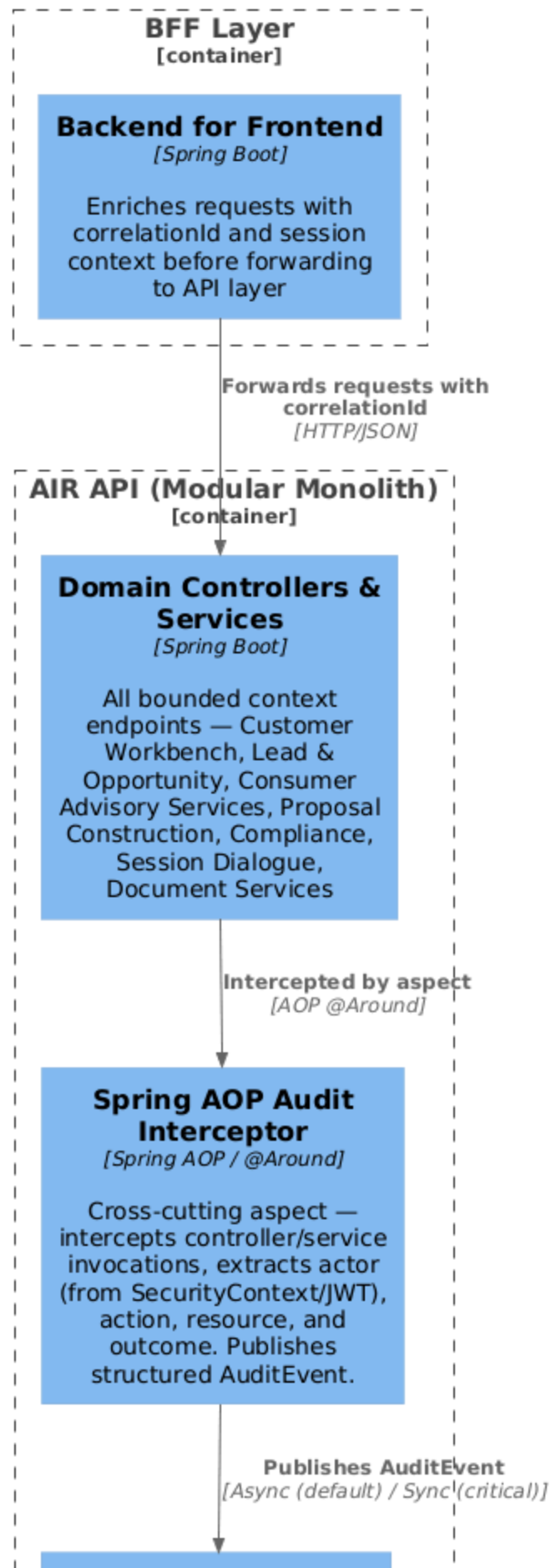
Cross-Cutting Concerns

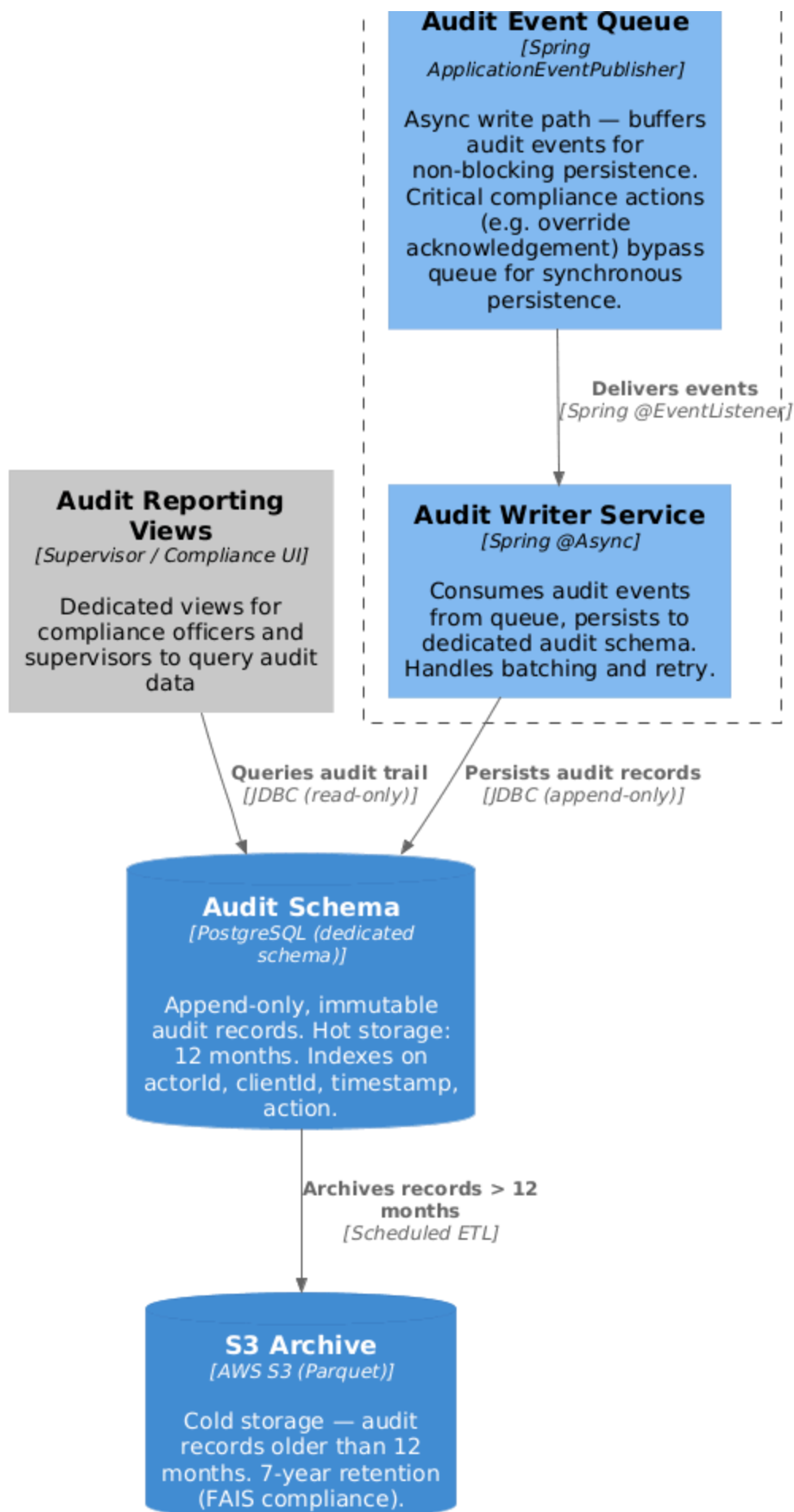
Application Interaction Audit Trail (ADR-036)

A cross-cutting audit service captures all significant user interactions across the AIR platform. The service is implemented via Spring AOP interceptors on controller/service boundaries, ensuring consistent capture without polluting domain logic.

Architecture:

Application Interaction Audit Trail — Cross-Cutting Architecture





Legend	
person	
system	
container	
component	
external person	
external system	
external container	
external component	

Audit Event Schema:

Field	Description
eventId	UUID
timestamp	ISO-8601 (UTC)
actorId	Entra ID user principal
actorRole	Role from JWT claims
action	Canonical action (e.g. VIEW_CLIENT_PROFILE , APPROVE_CASE)
module	Originating bounded context
resourceType	Entity type (e.g. AdviceCase , Proposal)
resourceId	Entity identifier
outcome	SUCCESS , DENIED , ERROR
metadata	JSON context (state changes, sections edited)
correlationId	Links related audit entries
clientId	Client identifier for per-client queries

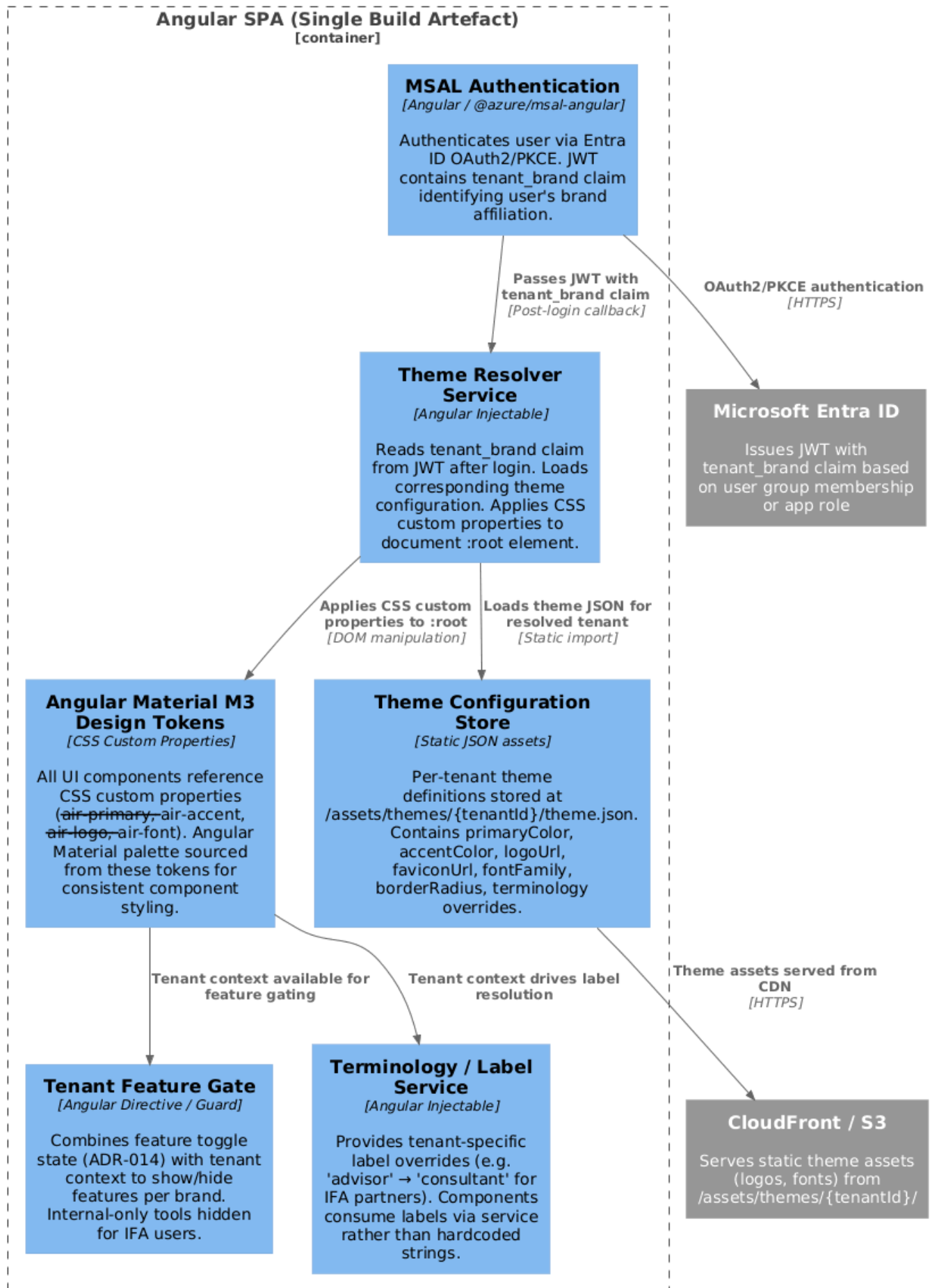
Auditable Action Categories: Data access, state transitions, proposal lifecycle, compliance interactions, AI agent interactions, lead management, consent & documents, administrative actions.

Tenant-Aware UI Theming (ADR-037)

The Angular SPA supports runtime brand skinning based on the authenticated user's tenant/brand affiliation. A single build artefact serves all brands – no per-tenant deployments.

Architecture:

Tenant-Aware UI Theming — Runtime Brand Skinning



Legend	
person	
system	
container	
component	
external person	
external system	
external container	
external component	

Key Design Decisions:

- Theme resolved from JWT claims at authentication – no additional API calls
- CSS custom properties as the theming layer – runtime switching without recompilation
- Angular Material M3 design tokens consume the custom properties
- Feature toggles (ADR-014) combine with tenant context for per-brand feature visibility
- Terminology overrides via i18n/label service (e.g. "advisor" → "consultant" for IFA partners)
- Default theme (FNB internal) applied immediately; tenant overrides applied post-authentication

Supported Tenants (MVP):

- `fnb-internal` – FNB-branded advisors (default)
- Future: `ifa-partner-*` – external IFA partners (white-label)

Diagram Summary

Level	Diagram	Purpose
C4 Level 1	System Context	Shows AIR in its ecosystem with users and external systems (including MFT, Avalon, Consent Engine)
C4 Level 2	Container	Shows deployable units – SPA, BFF, API monolith, Rules Engine, Agent Validation Service, DB, vector store, Bob & Vera agents, PII tokenisation, LLM gateway, event bus
C4 Level 3	Component (API)	Shows internal bounded context modules within the monolith (8 active contexts)
C4 Level 3	Component (Agents)	Shows Bob's cross-context read/write access, Vera's sync validation role, Agent Validation Service, external Rules Engine, PII tokenisation gateway, RAG pipeline, and LLM integration
C4 Level 4	Deployment	Shows AWS infrastructure topology – EKS (BFF, API, Rules Engine pods), RDS, CloudFront, S3, S3 Landing Zone (uses AWS Architecture Icons)

Level	Diagram	Purpose
Supplementary	CI/CD Pipeline	Shows trunk-based development with governance gates (acceptance criteria sign-off, test reports) and daily deployment flow
Supplementary	Event Flow	Shows event-driven orchestration with domain event catalogue
Supplementary	Cross-Cutting Concerns	Documents application audit trail and tenant-aware UI theming architecture

Key Architecture Characteristics

- **Modular Monolith:** All bounded contexts deployed as a single unit for speed, with explicit module boundaries for future decomposition
- **8 Active Bounded Contexts:** Customer Workbench, Lead & Opportunity Management, Customer Relationship & Insights, Consumer Advisory Services, Advisory Proposal Construction, Regulatory & Suitability Compliance, Session Dialogue & Contact, Document Services
- **Naming Convention:** All contexts use canonical service domain names where strong alignment to industry standards exists; bank-specific extensions are clearly documented
- **Backend for Frontend (BFF):** Java/Spring Boot API composition layer between SPA and monolith – aggregates granular domain APIs into coarser UI-optimised responses, handles AOP cross-cutting concerns. All SPA traffic routes through the BFF.
- **Data Source Integration (Three Patterns):** (1) Real-time APIs via PEPv3 for live client data and calculations, (2) Batch data-apis (AAD, Networth) refreshing 24-72 hours via PEPv3, (3) MFT file transfers (scorecard, SAS models) deposited to S3 landing zone for event-driven processing
- **Bob as Application-Layer Agent:** Cross-context personal assistant with read access to all modules and write access to Advisory Proposal Construction and Customer Workbench
- **Vera as Domain Service:** Synchronous compliance validation delegating deterministic checks to the external Rules Engine; uses RAG against a vector store (pgvector) to explain why compliance checks fail; async monitoring deferred
- **Agent Validation Service:** Separate container that validates all agent outputs before persistence – cross-checks Bob's numerical outputs against Avalon calculator, ensures Vera's decisions are consistent with Rules Engine determinations. Synchronous gate.
- **Rules Engine (External Container):** Externalised deterministic rule evaluation accessible by both the monolith and the AI agent layer. Rules evolve

independently of application code. Technology selection deferred (see ADR-023).

- **Gary Deferred:** Professional coach agent deferred to future phase (Servicing Activity Analysis context)
- **PII Tokenisation Gateway:** All agent prompts pass through a tokenisation layer that replaces personally identifiable information (names, ID numbers, account numbers, addresses) with reversible session-scoped tokens before submission to LLMs – responses are de-tokenised before returning to the domain layer
- **Vector Store (pgvector):** PostgreSQL with pgvector extension stores compliance knowledge embeddings – regulatory rules, precedent explanations, suitability guidance, and product rules – enabling RAG-based explanations for Vera and contextual support for Bob
- **LLM Gateway:** Unified interface to external LLM providers managing model routing, rate limiting, token budgets, retry/fallback strategies, and prompt audit logging
- **Lead & Opportunity Management (Reclassified):** Thin API layer – retrieves pre-scored leads from upstream data sources and captures advisor feedback. Scoring and prioritisation handled upstream in the data layer. Also houses notebook/dictation features for potential lead creation.
- **Opportunity Setup Wizard in Consumer Advisory Services:** Triggered by `OpportunityReadyForEngagement` event from Lead & Opportunity Management, owned by Consumer Advisory Services
- **Customer Onboarding for Advice:** Owned by Advisory Proposal Construction module (data/steps), orchestrated by Consumer Advisory Services. Scope to be detailed during analysis phase.
- **Consent Engine Integration:** External enterprise consent management system. Document Services module integrates via PEPv3 – checks existing consent before proceeding, captures new consent when required. No internal consent tracking.
- **Avalon Calculator:** 3rd-party advice calculation engine accessed via PEPv3. Serves as authoritative source for numerical validation of agent outputs.
- **Event-Driven Backbone:** 10 domain events orchestrate cross-module communication and agent behaviour
- **Policy Gates:** Critical state transitions (e.g. `CompleteProposal`) go through synchronous Vera validation
- **AI Cannot Change State:** Agents suggest, flag, and advise – humans approve and transition
- **CI/CD Governance:** Tickets require approved acceptance criteria before development (Ways of Working). Test reports generated at PR stage and release stage. Coverage gates enforced.
- **Feature Toggles:** Decouple code deployment from business release
- **Application Interaction Audit Trail:** Cross-cutting audit service captures all significant user interactions, state transitions, and data access events. Append-only immutable records with canonical schema (who, what, when, where, outcome). Implemented via Spring AOP interceptors with async write path.

Dedicated audit schema in PostgreSQL with 7-year retention (FAIS compliance). Supervisors and compliance officers access via dedicated reporting views. See ADR-036.

- **Tenant-Aware UI Theming:** Angular SPA supports runtime brand skinning based on authenticated user's tenant/brand affiliation (resolved from JWT claims). Single codebase/deployment serves multiple brands via CSS custom properties over Angular Material's M3 design token system. Enables internal FNB branding and future external IFA white-label presentations without separate builds. Theme configurations include colour palettes, logos, typography, and optional terminology overrides. Feature toggles combine with tenant context for per-brand feature visibility. See ADR-037.
- **PR Environments:** Ephemeral namespaces with mock data for rapid feedback
- **Module-Owned Data:** Each module manages its own schema via Liquibase

Glossary

Term	Definition
PEP	Policy Enforcement Point – the enterprise API gateway through which external service calls are routed (PEPv3 is the current version)
AIR	Advice & Intelligence Relationship – the platform being designed
BIAN	Banking Industry Architecture Network – industry-standard reference model for banking service domains
BFF	Backend for Frontend – API composition layer between SPA and domain APIs, handling aggregation and cross-cutting concerns
Bob	AI personal assistant agent – advisor's cross-context co-pilot (proposal drafting, queue reasoning, engagement prep, notebook interpretation)
Vera	AI compliance agent – synchronous rule validation, suitability enforcement, and RAG-powered compliance failure explanations
Gary	AI professional coach agent – behavioural analysis and coaching insights (deferred)
Avalon	3rd-party advice calculation engine – authoritative source for financial calculations, accessed via PEPv3
MFT	Managed File Transfer – enterprise file transfer mechanism delivering batch data (scorecard, SAS models) to S3 landing zone
iDNA	Internal data platform/data warehouse – source of batch datasets (AAD, Networth) exposed via data-apis

Term	Definition
Consent Engine	Enterprise consent management system – checks existing consent and captures new consent (accessed via PEPv3)
RoA	Record of Advice – regulatory documentation of advice given
FNA	Financial Needs Analysis – client assessment inputs
HPA	Horizontal Pod Autoscaler – Kubernetes auto-scaling mechanism
NBA	Next-Best-Action – the prioritised recommendation for what an advisor should do next (computed upstream in data layer)
QTD	Quarter-to-Date – performance measurement period
GAP	Pre-signature opportunity (not yet in pipeline)
Pipeline	Post-signature, pre-fulfilment opportunity (committed revenue)
CRM	Customer Relationship Management – service domain for managing customer relationships
RAG	Retrieval-Augmented Generation – technique that augments LLM prompts with relevant context retrieved from a vector store to improve accuracy and grounding
pgvector	PostgreSQL extension providing vector similarity search – used as the vector store for agent RAG
PII	Personally Identifiable Information – data that can identify an individual (names, ID numbers, account numbers, addresses)
PII Tokenisation	Process of replacing PII with reversible, session-scoped tokens before data leaves the trust boundary (i.e. before LLM submission)
LLM	Large Language Model – external AI model service used by agents for natural language generation and reasoning
Agent Validation Service	Separate container that validates agent outputs before persistence – ensures numerical accuracy and decision consistency
Rules Engine	Externalised deterministic rule evaluation container – accessible by both the monolith and AI agents
Application Audit Service	Cross-cutting service capturing all significant user interactions, state transitions, and data access events in an append-only immutable store for regulatory compliance (FAIS, FICA, POPIA)
Tenant-Aware Theming	Runtime UI skinning system that applies brand-specific visual identity (colours, logos, typography) based on the authenticated user's tenant claim, enabling multi-brand delivery from a single Angular SPA build
IFA	Independent Financial Advisor – external (non-FNB) advisor who may use AIR under a different brand/white-label presentation

Appendix A: BIAN v14 Service Domain Mapping

The AIR platform's bounded contexts are aligned to the Banking Industry Architecture Network (BIAN) v14 reference model. The table below maps each AIR domain service to its corresponding BIAN service domain(s).

Active Bounded Contexts

AIR Domain Service	Module	BIAN v14 Service Domain(s)	Classification	Notes
Customer Workbench	workbenchModule	Customer Workbench, Point of Service	Core	Owns queue presentation, scorecard, read project
Lead & Opportunity Management	leadOpportunityModule	Lead and Opportunity Management	Core (thin API)	Thin API to retrieve p leads, capt feedback, s lead creati Scoring/pr handled up houses notebook/d features.
Customer Relationship & Insights	customerInsightsModule	CRM, Party Reference Data Directory, Customer Financial Insights	Core (read model)	Read-only p consumes re APIs (PEPv data-apis (Networth), file data
Consumer Advisory Services	advisoryServicesModule	Consumer Advisory Services, Customer Case Management	Core	Owns stages transitions wizard, out anchors. Or onboarding creation pr

AIR Domain Service	Module	BIAN v14 Service Domain(s)	Classification	Notes
Advisory Proposal Construction	proposalModule	Consumer Advisory Services (proposal), Suitability Checking, Investment Portfolio Planning	Core	Owns proposal versioning, Owns customer onboarding (data/steps)
Regulatory & Suitability Compliance	complianceModule	Regulatory Compliance, Guideline Compliance, Suitability Checking	Core	Sync validation delegates to Rules Engine monitoring
Session Dialogue & Contact	sessionDialogueModule	Session Dialogue, Contact Handler, Servicing Event History	Supporting	Structured capture library
Document Services	documentServicesModule	Document Services, Correspondence	Supporting	Document generation compliance integrates external Co

Deferred Contexts (not represented in current architecture)

AIR Domain Service	BIAN v14 Service Domain(s)	Reason	Future Home
Party Authentication + Employee Access	Party Authentication, Employee Access	Consumed from bank IAM (Entra ID)	External dependency
External Advisor Authentication	Party Authentication	Non-FNB advisors lack Entra ID profiles; alternative identity federation needed	Future phase – approach TBD (see ADR deferred decisions)
Compliance Reporting	Compliance Reporting, Regulatory Reporting	Vera's async mode not yet validated	Future module + event subscriptions

AIR Domain Service	BIAN v14 Service Domain(s)	Reason	Future Home
Servicing Activity Analysis (Gary)	Servicing Activity Analysis, Employee Evaluation	Coach agent not in MVP scope	Future module + agent
Deal Execution & Fulfilment	<i>(Absorbed into Consumer Advisory Services)</i>	Terminal lifecycle stage	Extract when complexity warrants

Where AIR Extends Beyond BIAN v14

Extension	AIR Module	Rationale
AI Agent orchestration pattern	Bob, Vera, Gary	BIAN models capabilities, not cross-cutting agent constructs
Agent Validation Service	Agent Infrastructure	BIAN does not model AI output validation patterns
Externalised Rules Engine	Shared container	BIAN models compliance as a service domain, not the rule execution infrastructure
Event-driven backbone	Cross-cutting	BIAN models capabilities, not the event infrastructure connecting them
Notebook/dictation in Lead & Opportunity	Lead & Opportunity Management	BIAN excludes internal employee productivity tooling; housed here for lead creation workflow
Application Audit Service	Cross-cutting (all modules)	BIAN does not prescribe a unified application-level audit trail pattern; AIR implements this as a regulatory requirement (FAIS, FICA, POPIA)
Tenant-Aware UI Theming	Angular SPA (presentation)	BIAN does not model presentation-layer concerns; AIR requires multi-brand skinning for internal vs external advisor populations

Appendix B: Domain Event Catalogue

Complete catalogue of domain events published and consumed across the AIR platform's bounded contexts.

Published Events

Event Name	Publishing Module	Description	Delivery	Key Consumer
LeadFeedbackRecorded	Lead & Opportunity Management	Raised when an advisor provides feedback (accept, defer, dismiss) on a queued lead	Async	Bob (learn signal)
OpportunityReadyForEngagement	Lead & Opportunity Management	Raised when an opportunity has been fully qualified and is ready for the advisor to begin the engagement process	Async	Consumer Advisory Services (triggers Setup wizard)
AdviceCaseCreated	Consumer Advisory Services	Raised when a new advice case is instantiated from a qualified opportunity	Async	Customer Workbench (projecting Bob (preparing draft))
StageAdvanced	Consumer Advisory Services	Raised when an advice case transitions to the next lifecycle stage	Async	Customer Workbench (projecting Bob (content shift))
CaseCompleted	Consumer Advisory Services	Raised when an advice case reaches its terminal state (fulfilled or withdrawn)	Async	Customer Workbench (projecting Bob (content shift))
ProposalSectionEdited	Advisory Proposal Construction	Raised when an advisor (or Bob) edits a section of	Async	Bob (suggest content)

Event Name	Publishing Module	Description	Delivery	Key Consumer
		the advisory proposal		
ProposalCompleted	Advisory Proposal Construction	Raised when all required proposal sections are complete and the advisor submits for compliance validation	Sync	Regulatory Suitability Compliance (Vera validation gate)
EngagementCaptured	Session Dialogue & Contact	Raised when a client engagement session (call, meeting) is recorded and stored	Async	Bob (update context)
EngagementSummaryAvailable	Session Dialogue & Contact	Raised when a structured summary of a client engagement has been generated (transcript, objectives, outcomes)	Async	Bob (update context)
ClientAccepted	Document Services	Raised when a client formally accepts the proposed advice (signs RoA or equivalent consent via Consent Engine)	Async	Consumer Advisory Services (advance to fulfilment) Bob (prepare draft)
DocumentGenerated	Document Services	Raised when a final compliance artefact (PDF, RoA document)	Async	—

Event Name	Publishing Module	Description	Delivery	Key Consum
		has been generated		
ValidationCompleted	Regulatory & Suitability Compliance	Raised when Vera completes a compliance validation pass, carrying an allow or block outcome	Async	Consumer Advisory Services (gate resu

Subscriptions by Module

Consuming Module	Subscribed Events	Purpose
Customer Workbench	AdviceCaseCreated, StageAdvanced, CaseCompleted	Maintains the aggregated read projection for the advisor's active case dashboard
Consumer Advisory Services	OpportunityReadyForEngagement, ValidationCompleted, ClientAccepted	Triggers the Opportunity Setup wizard, processes compliance gate results, advances cases to fulfilment
Regulatory & Suitability Compliance	ProposalCompleted	Triggers synchronous Vera validation as a policy gate before case progression
Bob (AI Agent)	LeadFeedbackRecorded, AdviceCaseCreated, StageAdvanced, ProposalSectionEdited, EngagementCaptured, EngagementSummaryAvailable, ClientAccepted	Provides contextual awareness for proactive suggestions, draft generation, and engagement preparation

Consuming Module	Subscribed Events	Purpose
Vera (AI Agent)	ProposalCompleted	Triggers synchronous compliance validation and suitability checking